LABSHEET-09

Q1. Display the mounted file systems and their disk-space usage using df.

df

Q2. Display disk-space usage in human-readable format using df and identify the filesystem, total, used, available and mounted-on fields.

df -h

Q3. Display the size of the current directory and its contents using du.

du

Q4. Display directory sizes in human-readable format and identify which directories are consuming the most space.

du -h --max-depth=1 | sort -hr

Q5. Display the size of a selected directory including its subdirectories using du with an appropriate option.

du -sh /var/log/*

Q6. Display all available block devices, disks and partitions using lsblk.

lsblk

Q7. Use lsblk to identify the disk names, partition names, sizes, types and mount points on the system.

lsblk -o NAME,SIZE,TYPE,MOUNTPOINT

Q8. Display additional file-system information for block devices using an appropriate lsblk option.

lsblk -f

Q9. Identify the currently mounted file systems and their mount points using findmnt or an appropriate command.

findmnt

Q10. Create a dedicated directory named /mnt/labdisk to use as a mount point and verify its permissions and existence.

sudo mkdir -p /mnt/labdisk

ls -ld /mnt/labdisk

Q11. Attach a suitable test storage device or partition to /mnt/labdisk and verify that it is mounted correctly.

sudo mount /dev/sdb1 /mnt/labdisk

findmnt /mnt/labdisk

Q12. Display the disk-space usage of the mounted /mnt/labdisk file system using df.

df -h /mnt/labdisk

Q13. Create a test file inside /mnt/labdisk and verify that the file is stored on the mounted file system.

echo 'Test file content' | sudo tee /mnt/labdisk/testfile.txt

ls -l /mnt/labdisk

Q14. Unmount /mnt/labdisk using umount and verify that the file system is no longer mounted.

```
sudo umount /mnt/labdisk
findmnt /mnt/labdisk
```

Q15. Attempt to access a mounted file system that is currently in use, identify why unmounting may fail, and safely release the process before unmounting.

```
# Identify processes using the mount point:
sudo fuser -v /mnt/labdisk
# Kill active processes or exit the directory, then unmount:
sudo fuser -k /mnt/labdisk
sudo umount /mnt/labdisk
```

Q16. Display the partition table of a selected disk using an appropriate command and identify the partitions and their sizes.

```
sudo fdisk -l /dev/sdb
```

Q17. Create a practice partition on a dedicated test disk or virtual disk using an appropriate partitioning tool, then verify it with lsblk. Do not modify the system disk.

```
sudo fdisk /dev/sdb
# Press 'n' to create partition, 'p' for primary, set size, then 'w' to save
lsblk /dev/sdb
```

Q18. Create a file system on a dedicated practice partition, mount it on /mnt/labdisk and verify the file system using df, lsblk and findmnt.

```
sudo mkfs.ext4 /dev/sdb1
sudo mount /dev/sdb1 /mnt/labdisk
df -h /mnt/labdisk
lsblk /dev/sdb1
findmnt /mnt/labdisk
```

Q19. Unmount the practice file system and verify the mount point, device and partition information after unmounting.

```
sudo umount /mnt/labdisk
findmnt /mnt/labfs
lsblk /dev/sdb1
```

Q20. Perform a complete disk and storage management workflow: inspect disks and partitions, check disk usage, mount a dedicated test partition, create and verify data, unmount it and confirm the final storage state.

```
lsblk
df -h
sudo mkdir -p /mnt/labdisk
sudo mount /dev/sdb1 /mnt/labdisk
echo 'Disk Workflow Test' | sudo tee /mnt/labdisk/data.txt
ls -l /mnt/labdisk
sudo umount /mnt/labdisk
findmnt /mnt/labdisk
```